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A Family of Symmetric Distributions on the Circle

M. C. JONES and Arthur PEWSEY

We propose a new family of symmetric unimodal distributions on the circle that contains the uniform, von Mises, cardioid, and wrapped Cauchy distributions, among others, as special cases. The basic form of the densities of this family is very simple, although its normalization constant involves an associated Legendre function. The family of distributions can also be derived by conditioning and projecting certain bivariate spherically and elliptically symmetric distributions on to the circle. Trigonometric moments are available, and a measure of variation is discussed. Aspects of maximum likelihood estimation are considered, and likelihood is used to fit the family of distributions to an example set of data. Finally, extension to a family of rotationally symmetric distributions on the sphere is briefly made.

KEY WORDS: Associated Legendre function; Cardioid distribution; Circular uniform distribution; Distributions on the sphere; von Mises distribution; Wrapped Cauchy distribution.

1. INTRODUCTION

The von Mises distribution has a central role as a symmetric distribution on the circle. Other well-known symmetric distributions on the circle include the cardioid and wrapped Cauchy distributions. The circular uniform distribution is a special case of all of these. (See Mardia 1972; Batschelet 1981; Fisher 1993; Mardia and Jupp 1999; Jammalamadaka and SenGupta 2001 for book treatments of these distributions, to which the reader is referred for standard details.)

In this article we propose a simple general family of symmetric unimodal distributions on the circle that incorporates all of the aforementioned distributions. It has density

$$f_{\psi}(\theta) \propto \{1 + \tanh(\kappa\psi) \cos(\theta - \mu)\}^{1/\psi},$$
$$\mu - \pi < \theta \leq \mu + \pi. \quad (1)$$

Here μ is a location parameter, $\kappa \geq 0$ is a concentration parameter equivalent to the usual parameter of the von Mises distribution, and the index $\psi \in \Re$; the cases $\psi = 1, -1$, and, by continuity, 0 yield the cardioid, wrapped Cauchy, and von Mises distributions. Section 2 provides details on unimodality, normalizing constant, special cases, graphs, genesis of the distribution, trigonometric moments, and a measure of variation. Section 3.1 and the Appendix consider the asymptotics of maximum likelihood estimation, and Section 3.2 describes some practical aspects thereof. Section 4 then fits the family of distributions to an example of animal orientation data. Finally, Section 5 gives brief consideration to extending the family of symmetric circular distributions to a family of rotationally symmetric distributions on the sphere.

One important symmetric distribution on the circle that is not a member of family (1) is the wrapped normal distribution. It is well known that the von Mises and wrapped normal distributions match each other reasonably closely over the entire range of values for κ (Stephens 1963). We are aware of only one alternative three-parameter family of symmetric distributions on the circle, namely the wrapped symmetric stable dis-

tributions of Mardia (1972, sec. 3.4.8e), Jammalamadaka and SenGupta (2001, sec. 2.2.8), and Gatto and Jammalamadaka (2003). As their name suggests, these distributions arise by wrapping real-line stable distributions onto the circle. In general, their densities can be written only in terms of an infinite series, specifically as

$$f_{\alpha}(\theta) = \frac{1}{2\pi} \left\{ 1 + 2 \sum_{p=1}^{\infty} \rho^{p^{\alpha}} \cos(p(\theta - \mu)) \right\},$$
$$\mu - \pi < \theta \leq \mu + \pi,$$

where $\mu \in [0, 2\pi)$, $\rho \in [0, 1]$, and $\alpha \in (0, 2]$. The wrapped symmetric stable family and our new family are somewhat complementary. Their common special case is the wrapped Cauchy distribution. The new family includes the von Mises distribution, whereas the wrapped symmetric stable family includes the wrapped normal distribution. However, it can be argued that there are no direct analogs of the cardioid and other $\psi > 0$ distributions in the wrapped symmetric stable family, but then again, the wrapped normal can be rescaled to be similar to those distributions. More details on the comparison of (1) and wrapped stable distributions and other topics, such as circular kurtosis within the family and discrimination between von Mises, cardioid, and wrapped Cauchy distributions, along with more details of some of the claims made in what follows, have been given in earlier work (Jones and Pewsey 2004).

2. BASIC PROPERTIES

2.1 Unimodality and Identifiability

It can be easily shown that the density given by (1) is unimodal with mode at $\theta = \mu$ and antinode at $\theta = \mu + \pi$ for all $\psi \in \Re$ and $\kappa > 0$. When $\kappa = 0$, (1) gives the uniform density for all ψ . Note that in an obvious notation, $f_{\psi}(\theta; -\kappa, \mu) = f_{\psi}(\theta; \kappa, \mu + \pi)$. This distribution is the original distribution rotated to have mode at $\mu + \pi$; as is common practice with, for instance, the von Mises distribution, this duplication is the reason for restricting κ to be nonnegative throughout. In the remainder of Section 2 the location parameter μ is taken to be 0 without loss of generality.

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2.2 Normalizing Constant

Write $z = \cosh(\kappa\psi) \in [1, \infty)$, so that $f(\theta) \propto (z + \operatorname{sgn}(\psi) \times \sqrt{z^2 - 1} \cos \theta)^{1/\psi}$. Then, when $\psi > 0$, use the following standard integral from Gradshteyn and Ryzhik (1994, eq. 3.664.1):

$$\int_0^\pi (z + \sqrt{z^2 - 1} \cos x)^{1/\psi} dx = \pi P_{1/\psi}(z),$$

where $P_{1/\psi}(z)$ is the associated Legendre function of the first kind of degree $1/\psi$ and order 0 (Gradshteyn and Ryzhik 1994, secs. 8.7 and 8.8). For negative ψ ,

$$\begin{aligned} \int_{-\pi}^\pi \frac{1}{(z - \sqrt{z^2 - 1} \cos x)^{-1/\psi}} dx &= 2\pi P_{-(1/\psi)-1}(z) \\ &= 2\pi P_{1/\psi}(z), \end{aligned}$$

by (3.664.2) and (8.731.5) of Gradshteyn and Ryzhik. It follows that

$$f_\psi(\theta) = \frac{(\cosh(\kappa\psi) + \sinh(\kappa\psi) \cos \theta)^{1/\psi}}{2\pi P_{1/\psi}(\cosh(\kappa\psi))}. \quad (2)$$

2.3 Special Cases

Case 1: $\psi = 0$, the von Mises Distribution. The key is that for small $|\psi|$,

$$\begin{aligned} (\cosh(\kappa\psi) + \sinh(\kappa\psi) \cos \theta)^{1/\psi} &\simeq \exp\left\{\frac{1}{\psi} \log(1 + \kappa\psi \cos \theta)\right\} \\ &\simeq \exp(\kappa \cos \theta). \end{aligned}$$

Also, setting $\mu = 0$, $z = \kappa$, and $\nu = 1/\psi > 0$ in (9.6.49) of Abramowitz and Stegun (1965), $\lim_{\psi \downarrow 0} P_{1/\psi}(\cosh(\kappa\psi)) = I_0(\kappa)$, where I_0 is the modified Bessel function of the first kind of order 0. Essentially the same argument with $\nu = -1/\psi$ gives the same limit for negative ψ . So density (2) tends to the von Mises density $e^{\kappa \cos \theta} / (2\pi I_0(\kappa))$ as ψ tends to 0.

Case 2: $\psi = 1$, the Cardioid Distribution. $P_1(z) = z$, and so

$$f_1(\theta) = (2\pi)^{-1} (1 + 2\rho \cos \theta).$$

Here we have used the parameterization $0 \leq \rho = \tanh(\kappa/2) \leq 1/2$. Often, authors specify $|\rho| \leq 1/2$ (i.e., $\kappa \in \mathbb{R}$); we disallow negative ρ for the reason given in Section 2.1.

Case 3: $\psi = -1$, the Wrapped Cauchy Distribution. $P_{-1}(z) = P_0(z) = 1$, and so, in its usual parameterization,

$$f_{-1}(\theta) = \frac{1}{2\pi} \frac{1 - \rho^2}{1 + \rho^2 - 2\rho \cos(\theta)}, \quad 0 \leq \rho \leq 1,$$

where $\rho = \tanh(\kappa/2)$. (Along with the book treatments mentioned in Sec. 1, see McCullagh 1996 for further insights into wrapped Cauchy models.)

Case 4: $0 < \psi$, $\kappa \rightarrow \infty$, Cartwright's Power-of-Cosine Distribution. This has density

$$\begin{aligned} &\frac{2^{(1/\psi)-1} \Gamma^2((1/\psi) + 1) (1 + \cos(\theta))^{1/\psi}}{\pi \Gamma((2/\psi) + 1)} \\ &= \frac{2^{(2/\psi)-1} \Gamma^2((1/\psi) + 1) \cos(\theta/2)^{2/\psi}}{\pi \Gamma((2/\psi) + 1)} \quad (3) \end{aligned}$$

(Cartwright 1963). Note that Jammalamadaka and SenGupta (2001, sec. 2.2.2) made an unnecessary restriction to $\psi \leq 1$.

The normalizing constant—which has various alternative forms via properties of gamma and beta functions—follows because $\lim_{z \rightarrow \infty} z^{-1/\psi} P_{1/\psi}(z) = \Gamma((2/\psi) + 1) / \{2^{1/\psi} \Gamma^2((1/\psi) + 1)\}$ for $\psi > 0$ (eq. 8.776.1 of Gradshteyn and Ryzhik 1994). We also observe that Cartwright's distribution is that of $\Theta = 2 \sin^{-1}(B)$, where $B \sim \text{beta}(\alpha, \beta)$ on $[-1, 1]$ and $\alpha = \beta = (1/\psi) + (1/2)$ for $\psi > 0$. Such circular beta distributions for general α and β have been discussed by Jammalamadaka and SenGupta (2001, sec. 2.2.10), but the connection with Cartwright's distribution was not made there. Equivalently, $\Theta = 2 \tan^{-1}(T/\sqrt{\nu})$, where T follows the Student t distribution with ν degrees of freedom and $\nu = (2/\psi) + 1 \geq 1$, an observation also made by Minh and Farnum (2003).

Case 5: $-2 \leq \psi \leq 0$, $\kappa \rightarrow \infty$, Dirac Delta Function at 0. In these cases, it is of interest to obtain the limiting forms of (2) when appropriately standardized. For instance, setting $\xi_0 = \sqrt{\kappa}(\theta - \mu)$, Mardia and Jupp (1999, pp. 37–38) showed that the von Mises density behaves like a normal density for large κ . For $-2 < \psi < 0$, the appropriate standardization is $\xi_\psi = 2e^{\kappa\psi}(\theta - \mu)/\omega$, where $\omega = \sqrt{-(2/\psi) - 1}$. Then, using (8.776.1) of Gradshteyn and Ryzhik (1994) once again, further manipulation shows that the large- κ form of density (1) is the t density on ω degrees of freedom. We have no nondegenerate limit when $\psi = -2$.

Case 6: $-\infty < \psi < -2$, $\kappa \rightarrow \infty$, a New Distribution With a Pole at 0. Write $0 < \nu = -1/\psi < 1/2$ and $x = \kappa\psi > 0$, and use (8.776.1) of Gradshteyn and Ryzhik (1994) once more. Then $2\pi(\cosh x - \sinh x \cos \theta)^\nu P_{\nu-1}(\cosh x) \simeq \sqrt{\pi} \Gamma(-\nu + (1/2)) (1 - \tanh x \cos \theta)^\nu / \{2^{\nu-1} \Gamma(-\nu + 1)\}$. A little further manipulation allows the limiting density to be written just as in (3), except with the sine function replacing the cosine. Now this density is that of $\Theta = 2 \cos^{-1}(B)$, where $B \sim \text{beta}(\alpha, \beta)$ on $[-1, 1]$ and $\alpha = \beta = (1/\psi) + (1/2)$ for $0 > 1/\psi > -1/2$, and $\Theta = 2 \cot^{-1}(T/\sqrt{\nu})$, where $t \sim t_\nu$ and $0 < \nu = (2/\psi) + 1 < 1$. Notice that this completes the “missing” range of trigonometric transformations of beta and t distributions from Case 4.

Cases 7 and 8: $\kappa = 0$ and $\psi \rightarrow \pm\infty$, κ finite. Both yield the circular uniform distribution.

2.4 Graphs of Density

Density (2) is plotted in Figure 1 for $\psi = -1.5$ to 1.5 in steps of .5, in Figure 1(a) for $\kappa = 1/2$, and in Figure 1(b) for $\kappa = 2$. Note that these choices include standard distributions in the wrapped Cauchy when $\psi = -1$, von Mises when $\psi = 0$, and cardioid when $\psi = 1$. Considerable similarities are apparent between members of the family for the smaller value of κ ; these similarities increase as κ continues to become smaller, when all members of the family tend to the uniform distribution. That $\psi > 0$ produces a more spread distribution than $\psi \leq 0$ for the same value of κ is clear from Figure 1. For the larger value of κ , the $\psi < 0$ densities are starting to reflect their degenerate limit; positive ψ continues to yield well-spread distributions. There is relatively little difference between distributions with different values of ψ for small κ and between distributions with different values of κ for $\psi > 0$.

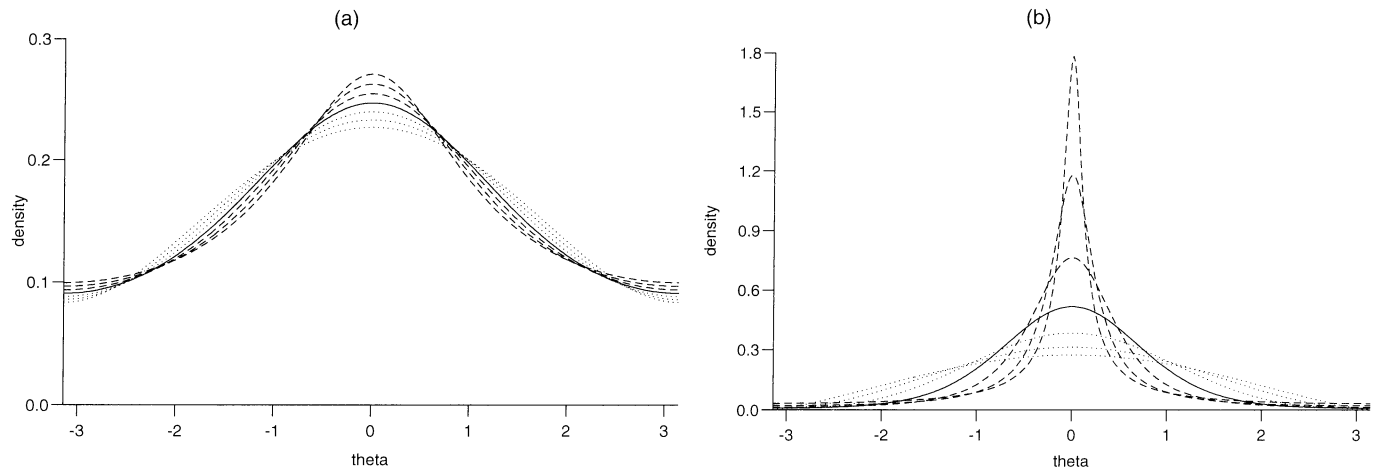


Figure 1. Density (2) for $\psi = -3/2, -1, -1/2$ (dashed; $\psi = -1$ is wrapped Cauchy), $\psi = 0$ (solid; von Mises), and $\psi = 1/2, 1, 3/2$ (dotted; $\psi = 1$ is cardioid), in Order of Decreasing Height at 0, for (a) $\kappa = 1/2$ and (b) $\kappa = 2$.

2.5 Conditional Derivations

Two different constructions involving conditioning bivariate symmetric distributions onto the unit circle lead to the same conditional distributions on the circle. The first concerns a spherically symmetric distribution with nonzero location, without loss of generality taken to be at $(m, 0)$, $m > 0$, that is, $(X_s, Y_s) \sim f_s(x, y) \propto g\{(x-m)^2 + y^2\}$ (Fang, Kotz, and Ng 1990, chap. 2). Then, the polar random variables $(R_s, \Theta_s) \sim r_s g\{(r_s^2 + m^2) - 2mr_s \cos \theta_s\}$, and thus, conditional on $R_s = r_s \in (0, \infty)$,

$$\Theta_s | R_s = r_s \sim g\{(r_s^2 + m^2) - 2mr_s \cos \theta_s\}. \quad (4)$$

Alternatively, let $(X_e, Y_e) \sim f_e(x, y) \propto g\{(x^2 - 2\delta xy + y^2)/(1 - \delta^2)\}$, $-1 \leq \delta < 1$, follow an elliptically symmetric distribution with mean at the origin (Fang et al. 1990, chap. 2). Then, letting (R_e, Φ) denote polar coordinates in this case,

$$\Phi | R_e = r_e \sim g\left\{\frac{r_e^2(1 - \delta \sin 2\phi)}{1 - \delta^2}\right\}.$$

This is of course an axial distribution, so double the angle to get $\Theta_e \equiv 2\Phi | R_e = r_e \sim g\{r_e^2(1 - \delta \sin \theta_e)/(1 - \delta^2)\}$. This has mode at $\pi/2$ if $\delta > 0$ and at $-\pi/2$ if $\delta < 0$. Recentering the distribution at 0, so that $\sin \theta_e$ transforms to $\sin\{\theta_e + \text{sgn}(\delta)\pi/2\} = \text{sgn}(\delta) \cos \theta_e$, we have

$$\Theta_e | R_e = r_e \sim g\left\{\frac{r_e^2(1 - |\delta| \cos \theta_e)}{1 - \delta^2}\right\}. \quad (5)$$

Formulas (4) and (5) match provided that we make the identifications $|\delta| = 2mr_s/(r_s^2 + m^2)$, $r_e = |r_s^2 - m^2|/\sqrt{r_s^2 + m^2}$ or, equivalently, $m = (r_e/2)d_+(\delta)$, $r_s = (r_e/2)d_-(\delta)$, where $d_{\pm}(\delta) = (1 - |\delta|)^{-1/2} \pm (1 + |\delta|)^{-1/2}$.

Three special choices for g cover virtually all of the family (1). As is well known, the von Mises distribution arises from conditioning on R in the bivariate normal distribution; $g(z) = e^{-z^2/2}$ yields $f_0(\theta) \propto e^{\kappa \cos \theta}$ where $0 < \kappa = mr_s = r_e^2|\delta|/[2(1 - \delta^2)] < \infty$. The bivariate t distribution (Fang et al. 1990, sec. 3.3) has $g(z) = (1 + z^2/v)^{-(v/2)-1}$, $v > 0$, which gives $f_\psi(\theta)$ for $0 \leq \tanh(\kappa\psi) = 2mr_s/(v + m^2 + r_s^2) = r_e^2|\delta|/[v(1 - \delta^2) + r_e^2] \leq 1$ and $0 > \psi > -1$. Conditioning a t distribution using the nonzero location spherically symmetric version, and hence this subset of our distributions, was considered briefly

by Shimizu and Iida (2002). The bivariate symmetric beta distribution (Fang et al. 1990, sec. 3.4) is based on $g(z) = (1 - z)^{b-(3/2)}I(0 \leq z \leq 1)$ for $b > 1/2$. The spherical version of this distribution centered at $(m, 0)$ completely envelops the circle of radius r_s centered at the origin only if $m + r_s \leq 1$. So we condition on $R_s = c_s(1 - m)$ for $0 \leq c_s \leq 1$. With $0 \leq \tanh(\kappa\psi) = 2mc_s/(1 + m - c_s^2(1 - m)) \leq 1$, for $b > 3/2$, we obtain $f_\psi(\theta)$ for $\psi > 0$, whereas values of $1/2 < b < 3/2$ provide members of family (1) with $-1 < \psi < \infty$, albeit with mode shifted from 0 to π . Likewise, the elliptical version of the beta distribution completely envelops only circles of radius $\sqrt{1 - |\delta|}$ or less, so we need to condition on $R_e^2 = c_e(1 - |\delta|)$ for $0 \leq c_e \leq 1$. The same values of ψ then ensue with $0 \leq \tanh(\kappa\psi) = c_e|\delta|/(1 - c_e + |\delta|) \leq 1$.

The only member of family (1) not available from this conditional derivation is the wrapped Cauchy distribution, $\psi = -1$. Interestingly, the wrapped Cauchy is the only member of the family that arises by projection onto, rather than conditioning on, the unit circle. It is the distribution obtained by doubling the angle in the axial distribution that is the angular central Gaussian distribution (Mardia and Jupp 1999, sec. 3.5.6). The latter is in fact obtained by projecting any bivariate elliptical distribution with mean at the origin, not just the Gaussian, onto the unit circle. As is also clear from this section of Mardia and Jupp, the wrapped Cauchy distribution does not arise from projecting a bivariate spherically symmetric distribution with nonzero mean onto the circle.

2.6 Trigonometric Moments and a Measure of Variation

For a symmetric circular density like f_ψ , $\phi_p = E(e^{ip\Theta}) = \alpha_p$, where $\{\phi_p : p = 0, \pm 1, \dots\}$ is the characteristic function of Θ and $\alpha_p = E(\cos(p\Theta))$ is the p th cosine moment. Using Gradshteyn and Ryzhik (1994, eq. 8.711.2), we find that

$$\phi_p = \begin{cases} \frac{\Gamma((1/\psi) + 1)P_{1/\psi}^p(\cosh(\kappa\psi))}{\Gamma((1/\psi) + p + 1)P_{1/\psi}(\cosh(\kappa\psi))}, & \psi > 0 \\ I_p(\kappa)/I_0(\kappa), & \psi = 0 \\ \frac{\Gamma((1/|\psi|) - p)P_{1/\psi}^p(\cosh(\kappa\psi))}{\Gamma(1/|\psi|)P_{1/\psi}(\cosh(\kappa\psi))}, & \psi < 0, \end{cases}$$

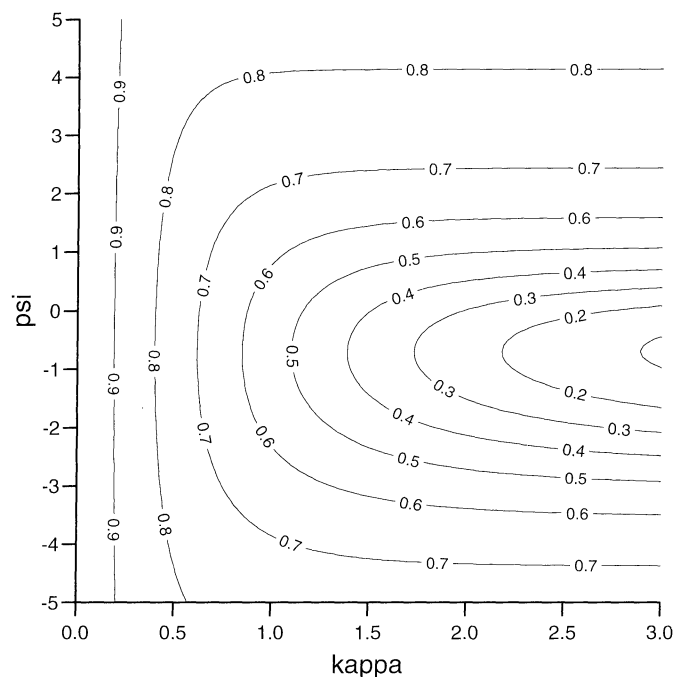


Figure 2. The Circular Variance as a Function of κ and ψ .

where $P_{1/\psi}^p(z)$ is the associated Legendre function of the first kind of degree $1/\psi$ and order p (Gradshteyn and Ryzhik 1994, secs. 8.7 and 8.8) and $I_p(\kappa)$ is the modified Bessel function of the first kind of order p . It can be confirmed that these general formulas reduce to standard formulas for $\psi = -1, 1$.

A common measure of variation on the circle is the circular variance $0 \leq v \leq 1$ given by $v = 1 - \rho$, where ρ denotes the mean resultant length. For a distribution with $\mu = 0$, as here, $\rho = \alpha_1$, which is given by

$$\alpha_1 = \frac{|\psi|}{(1 + \psi)} \frac{P_{1/\psi}^1(\cosh(\kappa\psi))}{P_{1/\psi}(\cosh(\kappa\psi))}.$$

The contour plot in Figure 2 represents the numerically calculated value of v as a function of κ and ψ . The circular variance takes its maximal value of 1 for the uniform distribution that corresponds to the left edge of Figure 2, at which $\kappa = 0$, and is also the limiting form to which the distributions are tending as $|\psi| \rightarrow \infty$. Note that the circular variance is not, as a first glance might suggest, symmetric in ψ . The circular variance decreases with increasing κ for all ψ , and this can be proved. As $\kappa \rightarrow \infty$, the circular variances for $\psi > 0$, $0 \geq \psi \geq -2$, and $\psi < -2$ tend to $\psi/(1 + \psi)$, 0 and $(2 + \psi)/(1 + \psi)$. A very similar contour plot arises if we instead consider another measure of variation on the circle, the circular dispersion of Fisher (1993, p. 42).

3. MAXIMUM LIKELIHOOD ESTIMATION

In general, our family of distributions has three parameters: location μ , concentration κ , and index ψ . Let $\theta_1, \dots, \theta_n$ be an independent and identically distributed sample assumed to be from one of these distributions. Write the log-likelihood as

$$\ell(\mu, \kappa, \psi) = \frac{1}{\psi} \sum_{i=1}^n \log(\cosh(\kappa\psi) + \sinh(\kappa\psi) \cos(\theta_i - \mu)) - n \log\{2Q_{1/\psi}(\kappa\psi)\},$$

where

$$Q_v(t) = \int_0^\pi (\cosh(t) + \sinh(t) \cos x)^v dx.$$

It will prove more convenient to work in terms of Q , its derivatives, and related functions rather than in terms of associated Legendre functions and their derivatives. All such functions are amenable to numerical integration, although care must be taken with numerical accuracy, particularly for large absolute values of $\kappa\psi$, and hence values of $\tanh(\kappa\psi)$ close to ± 1 .

3.1 Asymptotics

Standard asymptotic likelihood theory applies. For convenience, elements of the observed and expected information matrix for this model are given in the Appendix. First, note that the maximum likelihood estimate of location, μ , is asymptotically independent of the maximum likelihood estimates of the other two parameters, on which we now concentrate. Second, consideration of individual terms might seem to suggest that elements of the expected information matrix $\iota_{\kappa\psi} = O(1/\psi)$ and $\iota_{\psi\psi} = O(1/\psi^2)$ as $\psi \rightarrow 0$. However, it can readily be shown that the leading terms in both expressions cancel out, as does the $O(1/\psi)$ term of $\iota_{\psi\psi}$ after considerably more manipulation.

The asymptotic correlation between $\hat{\kappa}$ and $\hat{\psi}$ is given in terms of elements of the expected information matrix by $-\iota_{\kappa\psi}/(\iota_{\kappa\kappa}\iota_{\psi\psi} - \iota_{\kappa\psi}^2)$. A contour plot of this asymptotic correlation (not shown) gives the general impression of small to moderate correlations between $\hat{\kappa}$ and $\hat{\psi}$ for most κ and ψ , with the exception of higher correlations for small $|\psi|$, especially $\psi < 0$, and large κ . The asymptotic correlation is positive throughout. We have not found any simple reparameterization of (κ, ψ) to have smaller asymptotic correlations.

3.2 Practical Considerations

In practice, maximum likelihood estimation proceeds by numerical optimization of the log-likelihood function. The only difficulty is the one alluded to just before Section 3.1, that numerical instabilities in calculations arise when $|\kappa\psi|$ is large. In particular, we have imposed the restriction that $|\kappa\psi| < 6$ in our practical work. This has no bearing on the vast majority of situations, proving difficult only for distributions that either are close to being point distributions or are rather flat and cardioid-like. If the former is indicated for a particular dataset, then the data can be treated as being linear rather than circular, whereas in the latter case a cardioid distribution per se might be fitted.

We have found it useful to produce a contour plot of the (joint) profile likelihood surface of (κ, ψ) (with $|\kappa\psi| < 6$). Computation of this surface requires optimization over μ , and, as long as the distribution of the data is not close to uniform, the mean direction is a useful starting value; otherwise, it is advisable to use a range of starting values for μ spanning the whole of $(-\pi, \pi]$. Once a suitable range of κ and ψ values has been identified from an inspection of the profile log-likelihood surface, optimization of the log-likelihood is easily carried out using, for example, a grid-based direct-search algorithm incorporating the Nelder–Mead simplex algorithm (Nelder and Mead 1965). This was the approach that we used both here and in the initial “profiling” step.

In all of the analyses that we have performed, we found the profile log-likelihood surface (for $|\kappa\psi| < 6$) to be concave with a clearly identifiable maximum and contours around the maximum being close to elliptical in shape. This makes for rapid convergence to the maximum. An example for large n is given in the next section, but we have had the same experience with a range of smaller n datasets, although, as expected, the log-likelihood surface then becomes flatter about the maximum.

4. EXAMPLE

Here we consider a grouped dataset of $n = 714$ “vanishing angles” of nonmigratory British mallard ducks, taken from table 1.1 of either Mardia (1972) or Mardia and Jupp (1999) but originally given by Matthews (1961). Over a period of 1 year, the ducks were taken under sunny conditions from Slimbridge in Gloucestershire to various sites at distances of between 30 km and 250 km. Each duck’s vanishing angle is the bearing, measured clockwise from north, at which it disappeared from view when released. Mardia (1972, p. 10) described the distribution of the data as being “somewhat symmetrical,” and application of Pewsey’s (2002) test for reflective symmetry revealed no evidence indicating that the underlying distribution is not symmetric.

Results for fitting the full family of distributions and for fitting each of the von Mises, cardioid, and wrapped Cauchy submodels are given in Table 1. The von Mises distribution proves clearly the best of these three, but it is firmly rejected by a likelihood ratio test in favor of the member of the family that maximizes the likelihood, which has $\psi = -.35$. Indeed, approximate 95% confidence intervals for ψ are $(-.54, -.18)$ based on the profile log-likelihood and the χ^2_1 asymptotic distribution for the log-likelihood ratio, $(-.53, -.17)$ based on ± 1.96 times the standard error obtained from the observed information, and $(-.49, -.21)$ based on ± 1.96 times the standard error obtained from the expected information. The profile log-likelihood for (κ, ψ) (Fig. 3) is well behaved and suggests a single sensible maximum; the profile log-likelihood for ψ alone (not shown) is reasonably symmetric about $\psi = -.35$, which helps account for the similarity of these intervals.

The best-fitting member of the family of distributions and the fitted von Mises density are plotted together with a histogram of the data in Figure 4(a). The fit of the distribution with $\psi = -.35$ is noticeably better than that of the distribution with $\psi = 0$. A Q-Q plot of the ordered data $\theta_{(i)}$ against $F^{-1}((i - .5)/n)$ where F is the distribution function of the best-fitted model, is presented in Figure 4(b). Here F is defined by setting $F(140^\circ) = 0$, corresponding to the lower limit of the class interval closest to $\hat{\mu} - \pi$, which is also the left limit for θ in Figure 4(a); the coarseness of the Q-Q plot corresponds to

Table 1. Parameter Values Maximizing the Log-Likelihood and Corresponding Maximized Log-Likelihood Values for the Full-Distribution Family and Three Subfamilies for the Mallard Data

Distribution	ψ	μ	κ	Log-likelihood
Full family	-.35	-.81	1.84	-869.36
von Mises	0	-.80	2.12	-877.46
Cardioid	1	-.74	2.76	-981.55
Wrapped Cauchy	-1	-.82	1.50	-889.42

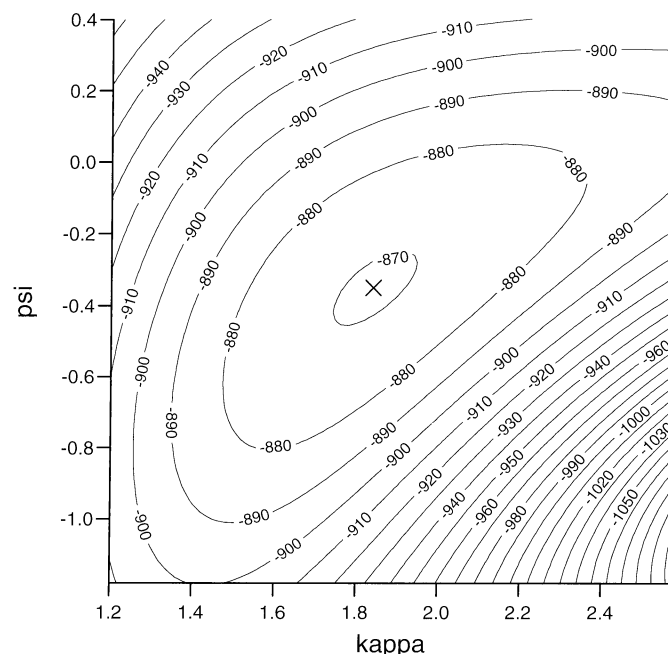


Figure 3. The Profile Log-Likelihood of κ and ψ for the Mallard Data. The position of the maximum likelihood estimates of κ and ψ is marked by a cross.

the grouping of the data. Chi-squared tests of the goodness of fit of these distributions, based on the binning shown in Figure 4(a) but combining class intervals to have expected values > 5 , emphatically reject the von Mises model ($p = .0002$) but are much more marginal about the $\psi = -.35$ model ($p = .0467$). The Q-Q plot perhaps makes more evident that any slight lack of fit might be due to the “left-hand tail” of the data being less heavy than that of the fitted model.

5. ROTATIONALLY SYMMETRIC DISTRIBUTIONS ON THE SPHERE

The immediate extension of family (1) to distributions on the sphere in $p \geq 2$ dimensions, S^{p-1} , has density $f_\psi(\mathbf{x}) \propto \{1 + \tanh(\kappa\psi)\mathbf{x}^T\boldsymbol{\mu}\}^{1/\psi}$, where $\mathbf{x}$ and $\boldsymbol{\mu}$ are unit vectors and superscript “ T ” denotes vector transpose. As such, the distributions are apparently novel special cases of the general family of spherical models with rotational symmetry (about $\boldsymbol{\mu}$) (see, e.g., Watson 1983, chap. 4; Mardia and Jupp 1999, sec. 9.3.3). The preexisting special case is the von Mises–Fisher–Langevin distribution (Mardia and Jupp 1999, sec. 9.3.2) corresponding to $\psi = 0$.

Following Watson and Mardia and Jupp, rotationally symmetric spherical random variables can be written as $\mathbf{x} = T\boldsymbol{\mu} + \sqrt{1 - T^2}\boldsymbol{\xi}$, where $\boldsymbol{\xi}$ is the unit tangent to S^{p-1} at $\boldsymbol{\mu}$, uniformly distributed on S^{p-2} and independent of the scalar random variable $T = \mathbf{x}^T\boldsymbol{\mu}$. When $\mathbf{x}$ is uniform on S^{p-1} , T has density $B((p-1)/2, 1/2)^{-1}(1-t^2)^{(p-3)/2}$ on $t \in [-1, 1]$, where $B(\cdot, \cdot)$ denotes the beta function. It follows that

$$\begin{aligned} \int_{S^{p-1}} (1 + \tanh(\kappa\psi)\mathbf{x}^T\boldsymbol{\mu})^{1/\psi} d\mathbf{x} \\ = B((p-1)/2, 1/2)^{-1} \\ \times \int_{-1}^1 (1 + \tanh(\kappa\psi)t)^{1/\psi} (1-t^2)^{(p-3)/2} dt. \end{aligned}$$

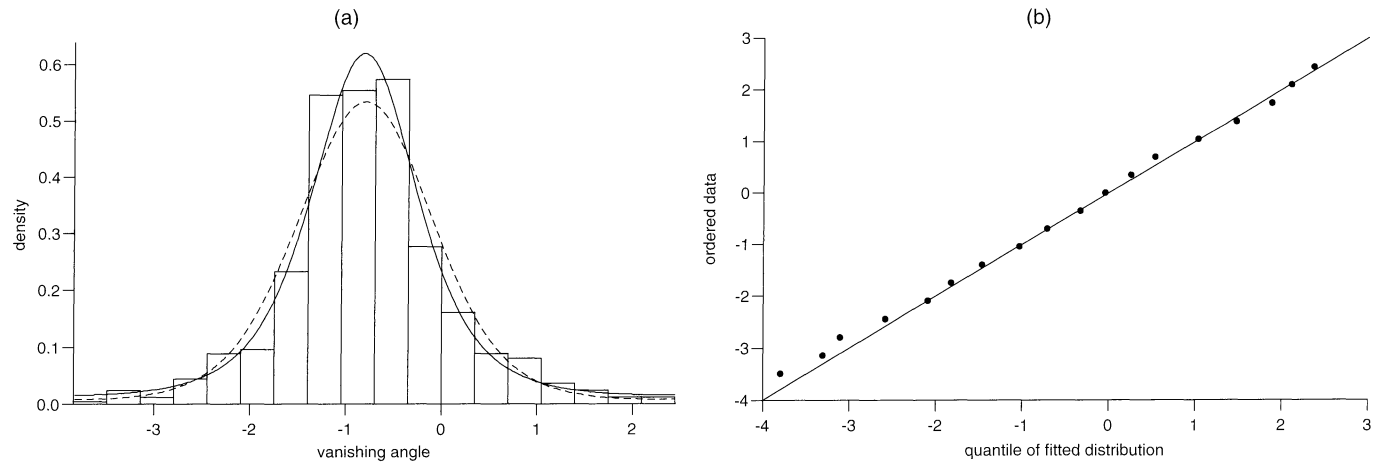


Figure 4. The Mallard Data and Fitted Distributions. (a) Histogram of the data together with maximum likelihood fits of the full family of distributions (solid curve) and the von Mises distribution (dashed curve). The data and densities are plotted on approximately $(\hat{\mu} - \pi, \hat{\mu} + \pi)$. (b) Q-Q plot of the data and the best-fitting distribution from the full family. The distribution function is calculated with respect to the range for θ defined in (a).

This can be evaluated for ψ both positive and negative by judicious use of (8.711.1) of Gradshteyn and Ryzhik (1994). For $\psi \neq 0$, the density on the sphere with respect to the uniform distribution turns out to be

$$f_{\psi}(\mathbf{x}) = \frac{|\sinh(\kappa\psi)|^{(p/2)-1}}{2^{(p/2)-1}\Gamma(p/2)} \frac{(\cosh(\kappa\psi) + \sinh(\kappa\psi)\mathbf{x}^T\boldsymbol{\mu})^{1/\psi}}{P^{1-(p/2)}_{(1/\psi)+(p/2)-1}(\cosh(\kappa\psi))}.$$

This reduces to $2\pi \times (2)$ when $p = 2$ and also reduces to a simpler form when $p = 3$, namely

$$f_{\psi}(\mathbf{x}) = \frac{(1 + \psi) \sinh(\kappa\psi)}{\psi \sinh(\kappa(1 + \psi))} (\cosh(\kappa\psi) + \sinh(\kappa\psi)\mathbf{x}^T\boldsymbol{\mu})^{1/\psi}$$

if $\psi \neq -1$, with another simple formula for $\psi = -1$. The nonzero mean spherically symmetric conditioning arguments of Section 2.5 can be extended, as has been shown in special cases by Downs (1966) and Shimizu and Iida (2002).

APPENDIX: ELEMENTS OF INFORMATION MATRICES

Here primes denote differentiation of $Q_{\nu}(t)$ with respect to $\cosh(t)$, and superscript circles denote differentiation of $Q_{\nu}(t)$ with respect to ν .

A.1 Observed Information

Elements of the observed information matrix are minus the second derivatives of ℓ ; denote them by J with subscripts $\mu\mu$, $\mu\kappa$, and so on. We have the following:

$$J_{\mu\mu} = \frac{1}{\psi} \sinh(\kappa\psi) \sum_{i=1}^n \frac{\sinh(\kappa\psi) + \cosh(\kappa\psi) \cos(\theta_i - \mu)}{(\cosh(\kappa\psi) + \sinh(\kappa\psi) \cos(\theta_i - \mu))^2},$$

$$J_{\mu\kappa} = - \sum_{i=1}^n \frac{\sin(\theta_i - \mu)}{(\cosh(\kappa\psi) + \sinh(\kappa\psi) \cos(\theta_i - \mu))^2},$$

$$J_{\mu\psi} = \frac{\kappa}{\psi} J_{\mu\kappa} + \frac{1}{\psi^2} \sinh(\kappa\psi) \sum_{i=1}^n \frac{\sin(\theta_i - \mu)}{\cosh(\kappa\psi) + \sinh(\kappa\psi) \cos(\theta_i - \mu)},$$

$$J_{\kappa\kappa} = -\psi \sum_{i=1}^n \frac{1 - \cos^2(\theta_i - \mu)}{(\cosh(\kappa\psi) + \sinh(\kappa\psi) \cos(\theta_i - \mu))^2} + n\psi^2 \left\{ \cosh(\kappa\psi) \frac{Q'_{1/\psi}(\kappa\psi)}{Q_{1/\psi}(\kappa\psi)} + \sinh^2(\kappa\psi) \left(\frac{Q''_{1/\psi}(\kappa\psi)}{Q_{1/\psi}(\kappa\psi)} - \frac{(Q'_{1/\psi}(\kappa\psi))^2}{Q_{1/\psi}^2(\kappa\psi)} \right) \right\},$$

$$J_{\kappa\psi} = -\kappa \sum_{i=1}^n \frac{1 - \cos^2(\theta_i - \mu)}{(\cosh(\kappa\psi) + \sinh(\kappa\psi) \cos(\theta_i - \mu))^2} + n\{\sinh(\kappa\psi) + \kappa\psi \cosh(\kappa\psi)\} \frac{Q'_{1/\psi}(\kappa\psi)}{Q_{1/\psi}(\kappa\psi)} + n\kappa\psi \sinh^2(\kappa\psi) \left\{ \frac{Q''_{1/\psi}(\kappa\psi)}{Q_{1/\psi}(\kappa\psi)} - \frac{(Q'_{1/\psi}(\kappa\psi))^2}{Q_{1/\psi}^2(\kappa\psi)} \right\} - \frac{n}{\psi} \sinh(\kappa\psi) \left\{ \frac{Q'^{\circ}_{1/\psi}(\kappa\psi)}{Q_{1/\psi}(\kappa\psi)} - \frac{Q'_{1/\psi}(\kappa\psi) Q_{1/\psi}^{\circ}(\kappa\psi)}{Q_{1/\psi}^2(\kappa\psi)} \right\},$$

and

$$J_{\psi\psi} = \frac{2\kappa}{\psi^2} \sum_{i=1}^n \frac{\sinh(\kappa\psi) + \cosh(\kappa\psi) \cos(\theta_i - \mu)}{\cosh(\kappa\psi) + \sinh(\kappa\psi) \cos(\theta_i - \mu)} - \frac{\kappa^2}{\psi} \sum_{i=1}^n \frac{1 - \cos^2(\theta_i - \mu)}{(\cosh(\kappa\psi) + \sinh(\kappa\psi) \cos(\theta_i - \mu))^2} + n\kappa^2 \sinh^2(\kappa\psi) \left\{ \frac{Q''_{1/\psi}(\kappa\psi)}{Q_{1/\psi}(\kappa\psi)} - \frac{(Q'_{1/\psi}(\kappa\psi))^2}{Q_{1/\psi}^2(\kappa\psi)} \right\} - \frac{2n\kappa}{\psi^2} \sinh(\kappa\psi) \left\{ \frac{Q'^{\circ}_{1/\psi}(\kappa\psi)}{Q_{1/\psi}(\kappa\psi)} - \frac{Q'_{1/\psi}(\kappa\psi) Q_{1/\psi}^{\circ}(\kappa\psi)}{Q_{1/\psi}^2(\kappa\psi)} \right\} + \frac{n}{\psi^4} \left\{ \frac{Q_{1/\psi}^{\circ\circ}(\kappa\psi)}{Q_{1/\psi}(\kappa\psi)} - \frac{(Q_{1/\psi}^{\circ}(\kappa\psi))^2}{Q_{1/\psi}^2(\kappa\psi)} \right\} + n\kappa^2 \cosh(\kappa\psi) \frac{Q'_{1/\psi}(\kappa\psi)}{Q_{1/\psi}(\kappa\psi)}$$

$$-\frac{2}{\psi^3} \left\{ \sum_{i=1}^n \log(\cosh(\kappa\psi) + \sinh(\kappa\psi) \cos(\theta_i - \mu)) - n \frac{Q_{1/\psi}^{\circ}(\kappa\psi)}{Q_{1/\psi}(\kappa\psi)} \right\}.$$

Here

$$Q'_v(t) = v \int_0^\pi (\cosh(t) + \sinh(t) \cos x)^{v-1} (1 + \coth(t) \cos x) dx,$$

$$Q_v^{\circ}(t) = \int_0^\pi (\cosh(t) + \sinh(t) \cos x)^v \times \log(\cosh(t) + \sinh(t) \cos x) dx,$$

$$Q''_v(t) = v(v-1) \int_0^\pi (\cosh(t) + \sinh(t) \cos x)^{v-2} \times (1 + \coth(t) \cos x)^2 dx$$

$$- v \sinh^{-3}(t) \int_0^\pi (\cosh(t) + \sinh(t) \cos x)^{v-1} \cos x dx,$$

$$Q_v^{\circ\circ}(t) = \int_0^\pi (\cosh(t) + \sinh(t) \cos x)^{v-1} (1 + \coth(t) \cos x) dx$$

$$+ v \int_0^\pi (\cosh(t) + \sinh(t) \cos x)^{v-1} (1 + \coth(t) \cos x) \times \log(\cosh(t) + \sinh(t) \cos x) dx,$$

and

$$Q_v^{\circ\circ}(t) = \int_0^\pi (\cosh(t) + \sinh(t) \cos x)^v \times \log^2(\cosh(t) + \sinh(t) \cos x) dx.$$

A.2 Expected Information

Elements of the expected information matrix follow from those of the observed information matrix. Denote n^{-1} times the expected information matrix by ι . Then

$$\iota_{\mu\mu} = \frac{1}{\psi} \sinh(\kappa\psi) \frac{R_{1/\psi}(\kappa\psi)}{Q_{1/\psi}(\kappa\psi)}, \quad \iota_{\mu\kappa} = \iota_{\kappa\mu} = 0,$$

$$\iota_{\kappa\kappa} = -\psi \frac{S_{1/\psi}(\kappa\psi)}{Q_{1/\psi}(\kappa\psi)} + \psi^2 \left\{ \cosh(\kappa\psi) \frac{Q'_{1/\psi}(\kappa\psi)}{Q_{1/\psi}(\kappa\psi)} + \sinh^2(\kappa\psi) \left(\frac{Q''_{1/\psi}(\kappa\psi)}{Q_{1/\psi}(\kappa\psi)} - \frac{(Q'_{1/\psi}(\kappa\psi))^2}{Q_{1/\psi}^2(\kappa\psi)} \right) \right\},$$

$$\iota_{\kappa\psi} = -\kappa \frac{S_{1/\psi}(\kappa\psi)}{Q_{1/\psi}(\kappa\psi)} + \{\sinh(\kappa\psi) + \kappa\psi \cosh(\kappa\psi)\} \frac{Q'_{1/\psi}(\kappa\psi)}{Q_{1/\psi}(\kappa\psi)} + \kappa\psi \sinh^2(\kappa\psi) \left\{ \frac{Q''_{1/\psi}(\kappa\psi)}{Q_{1/\psi}(\kappa\psi)} - \frac{(Q'_{1/\psi}(\kappa\psi))^2}{Q_{1/\psi}^2(\kappa\psi)} \right\}$$

$$- \frac{1}{\psi} \sinh(\kappa\psi) \left\{ \frac{Q_{1/\psi}^{\circ}(\kappa\psi)}{Q_{1/\psi}(\kappa\psi)} - \frac{Q'_{1/\psi}(\kappa\psi) Q_{1/\psi}^{\circ}(\kappa\psi)}{Q_{1/\psi}^2(\kappa\psi)} \right\},$$

and

$$\iota_{\psi\psi} = \frac{2\kappa}{\psi^2} \frac{R_{1/\psi}(\kappa\psi)}{Q_{1/\psi}(\kappa\psi)} - \frac{\kappa^2}{\psi} \frac{S_{1/\psi}(\kappa\psi)}{Q_{1/\psi}(\kappa\psi)} + \kappa^2 \sinh^2(\kappa\psi) \left\{ \frac{Q''_{1/\psi}(\kappa\psi)}{Q_{1/\psi}(\kappa\psi)} - \frac{(Q'_{1/\psi}(\kappa\psi))^2}{Q_{1/\psi}^2(\kappa\psi)} \right\}$$

$$- \frac{2\kappa}{\psi^2} \sinh(\kappa\psi) \left\{ \frac{Q_{1/\psi}^{\circ}(\kappa\psi)}{Q_{1/\psi}(\kappa\psi)} - \frac{Q'_{1/\psi}(\kappa\psi) Q_{1/\psi}^{\circ}(\kappa\psi)}{Q_{1/\psi}^2(\kappa\psi)} \right\}$$

$$+ \frac{1}{\psi^4} \left\{ \frac{Q_{1/\psi}^{\circ\circ}(\kappa\psi)}{Q_{1/\psi}(\kappa\psi)} - \frac{(Q_{1/\psi}^{\circ}(\kappa\psi))^2}{Q_{1/\psi}^2(\kappa\psi)} \right\}$$

$$+ \kappa^2 \cosh(\kappa\psi) \frac{Q'_{1/\psi}(\kappa\psi)}{Q_{1/\psi}(\kappa\psi)}.$$

Here

$$R_v(t) = \int_0^\pi (\cosh(t) + \sinh(t) \cos x)^{v-1} (\sinh(t) + \cosh(t) \cos x) dx$$

and

$$S_v(z) = \int_0^\pi (\cosh(t) + \sinh(t) \cos x)^{v-2} (1 - \cos^2 x) dx.$$

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